The empirical recurrence for OEIS A269586, proved from an exact model

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13 September 2026

Abstract

OEIS A269586 carries an empirical recurrence of order 7. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by an exact polynomial in n past a computed transient, from which a provably satisfies a MONIC linear recurrence of order $S = 26$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

2020 Mathematics Subject Classification. 05A15, 05A19, 11P21.

1 The conjecture

OEIS A269586 is "Number of length-6 0..n arrays with no repeated value differing from the previous repeated value by more than one.". It has offset 1 and begins

$$64, 659, 3680, 14239, 43184, 110339, 248464, 507935, \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = n^6 + 6*n^5 + 9*n^4 + 22*n^3 + 15*n^2 + 12*n - 1. a(n) = 7*a(n-1) - 21*a(n-2) + 35*a(n-3) - 35*a(n-4) + 21*a(n-5) - 7*a(n-6) + a(n-7) for n>7.

As of the "Last modified" line on the live entry (Jan 24 2019 11:45:32, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a polynomial past a computed transient

A REPEATED VALUE is a term equal to the one before it, and its value is that term; the entry's condition relates each repeated value to the previous one. The length is fixed at 6 and the alphabet $0..n$ grows, so nothing here is a walk on a finite digraph — the state that remembers the previous repeated value has as many values as the alphabet. The count is taken exactly instead. Writing $A[p][v]$ for the number of prefixes ending in v whose previous repeated value is p, appending a term t gives

$$A'[p][t] \mathrel{+}= \Big(\sum_v A[p][v]\Big) - A[p][t], \qquad A'[t][t] \mathrel{+}= A[p][t] \text{ when the condition allows,}$$

the first for a t differing from the last term and the second for a t equal to it. Every term the model produces is an exact integer.

3 The bound

Let k be the point past which the condition stops caring: the predicate depends on the difference of the two repeated values only through its sign and its magnitude up to k, and is constant beyond that in each direction. Here $k = 1$, read off the predicate rather than assumed. An array is determined by its weak ordering — the ordered set partition of the 6 positions into m blocks of equal value, listed in increasing value — together with the base value $g_0 \geq 0$ and the gaps $g_1, \ldots, g_{m-1} \geq 1$ between consecutive distinct values, subject to $g_0 + \sum g_i \leq n$. Each constraint is on a difference of two values, which is a signed sum of consecutive gaps; since the predicate is constant past k, each constraint splits into finitely many cases, and in every case it either fixes that gap-sum to one of at most $2k+1$ values or pushes it beyond k. Within a case the count is the number of integer points of a system (fixed amounts) plus (free gaps) $\leq n$, a polynomial in n of degree at most 6, valid as soon as n exceeds the total forced amount. That total is at most $(m-1) + (\text{constrained pairs})(k+1) \leq L + L(k+1)$, so with

$$n_0 \;=\; L + L(k+1) \;=\; 18$$

the sequence agrees with a polynomial of degree at most 6 for every $n \geq n_0$. It is therefore annihilated by $(z-1)^{6+1}$ from that point, and by $z^{n_0+1}(z-1)^{6+1}$ everywhere, a monic recurrence of order $S = 26$. The annihilator was then asked for terms the model had not supplied and reproduced them, which is what a bound one short fails.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j<S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^7 - \sum_i c_i z^{7-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a, so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a, hence every shift of a, hence any linear combination of shifts; that is r. A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. □

Theorem 2.

$$a(n) \;=\; 7\,a(n-1) - 21\,a(n-2) + 35\,a(n-3) - 35\,a(n-4) + 21\,a(n-5) - 7\,a(n-6) + a(n-7)$$

for every $n > 7$.

Proof. The terms $a(0), \ldots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 7 + 1$ onwards, and the run exceeds $S + 7$, which by Lemma 1 settles every larger index at once. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 26 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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[4] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Springer GTM 228, 2005.